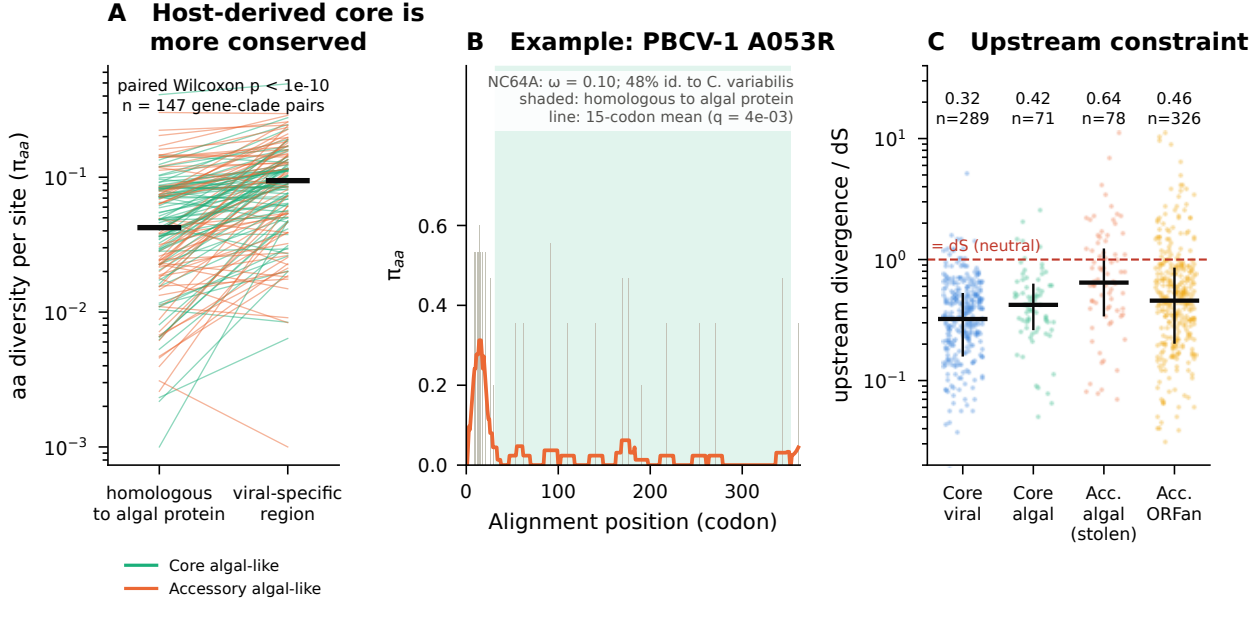
A Host-derived core is more conserved
paired Wilcoxon p < 1e-10
n = 147 gene-clade pairs
aa diversity per site (π_{aa})
10^{-1}
10^{-2}
10^{-3}
homologous to algal protein
viral-specific region
B Example: PBCV-1 A053R
NC64A: ω = 0.10; 48% id. to C. variabilis
shaded: homologous to algal protein
line: 15-codon mean (q = 4e-03)
π_{aa}
0.6
0.4
0.2
0.0
0
100
200
300
Alignment position (codon)
C Upstream constraint
upstream divergence / dS
0.32
n=289
0.42
n=71
0.64
n=78
0.46
n=326
10^{1}
10^{0}
10^{-1}
= dS (neutral)
Core viral
Core algal
Acc. algal (stolen)
Acc. ORFan
Core algal-like
Accessory algal-like